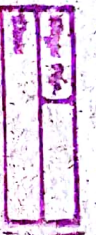


Unit-2



Laplace Transformations

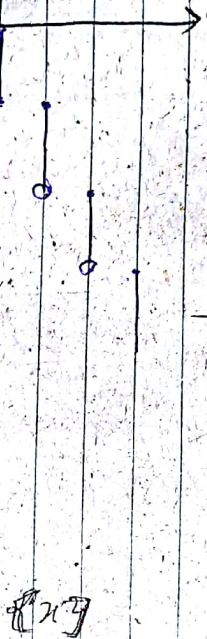
$$\mathcal{L}[f(t)] \equiv F(p) \quad \text{Parameter } p > 0$$

$$= \int_0^{\infty} f(t) e^{-pt} dt \quad \text{Kernel of LT}$$

Kernel is zero $\forall t < 0$

Existence Theory Th^m

ii) $f(t)$ should be piecewise continuous



In every finite interval in range $t > 0$

ii) $f(t)$ should be an exponential funcⁿ

such that $|f(t)| \leq M e^{at}$ $\forall t > 0$

$\Rightarrow$ expⁿ is bounded above funcⁿ



If both condⁿ exist then $\mathcal{L}[f(t)] = F(p)$ exists

This existence th^m is sufficient condⁿ for the existence of Laplace Trans^l but not necessarily

eg: $f(t) = \frac{1}{\sqrt{t}}$

i) $\lim_{t \rightarrow 0^+} f(t) = \frac{1}{\sqrt{t}} \rightarrow \infty$

Not a piecewise continuous in any interval which involves 0.

ii) Choose $M = 1$ & $a = 0$

$$|f(t)| = \left| \frac{1}{\sqrt{t}} \right| \leq M e^{at} \leq 1 \quad \forall t > 0$$

This is correct.

$\phi(t) = \frac{1}{\sqrt{t}}$ has a Laplace Transform $\forall p > a$

$$L\left[\frac{1}{\sqrt{t}}\right] = \int_0^{\infty} \frac{1}{\sqrt{t}} e^{-pt} dt$$

$$pt = x^2$$

$$p dt = 2x dx$$

$$\int_0^{\infty} \frac{1}{x\sqrt{p}} e^{-x^2} 2x dx$$

$$\frac{2}{\sqrt{p}} \int_0^{\infty} e^{-x^2} dx$$

$$= \frac{2}{\sqrt{p}} \left[\frac{\sqrt{x}}{2} \right]_0^{\infty} = \frac{\sqrt{x}}{p}$$

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{x}}{2}$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{x}$$

$$L\left[\frac{1}{\sqrt{t}}\right] = \sqrt{p}$$

Standard func

$$i) L[1] = \int_0^{\infty} e^{-pt} dt (1)$$

$$= \frac{1}{p} [e^{-pt}]_0^{\infty}$$

$$L[1] = \frac{1}{p}, \forall p > 0$$

$$ii) L[e^{at}] = \int_0^{\infty} e^{at} e^{-pt} dt$$

$$= \int_0^{\infty} e^{-(p-a)t} dt$$

$$= \frac{1}{-(p-a)} [e^{-(p-a)t}]_0^{\infty}$$

$$L[e^{at}] = \frac{1}{p-a}, \forall p > a$$

iii. $\mathcal{L}[t^n] = \int_0^{\infty} t^n e^{-pt} dt$

$$\int_0^{\infty} e^{-an} n^{m-1} dn = \left[\frac{n^m}{a^m} \right]_0^{\infty}$$

$$\int_0^{\infty} e^{-pt} t^{(n+1)-1} dt$$

$$\mathcal{L}[t^n] = \frac{n!}{p^{n+1}} \quad \forall p > 0 \quad \forall n > 0$$

if n is a +ve integer.

$$\int n+1 = \ln$$

$$\mathcal{L}[t^n] = \frac{n!}{p^{n+1}} \quad \forall p > 0 \quad \forall n > 0 \quad \forall n \in \mathbb{Z}$$

iv. $\mathcal{L}[\sin at] = \int_0^{\infty} \sin at e^{-pt} dt$

Reduceⁿ formula

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$= \frac{e^{-px}}{p^2 + a^2} \left[-p \sin ax - a \cos ax \right]_0^{\infty}$$

$$= \frac{-1}{p^2 + a^2} [-a]$$

$$\mathcal{L}[\sin at] = \frac{a}{p^2 + a^2} \quad \forall p > 0$$

$$\mathcal{L}[\cos at] = \frac{p}{p^2 + a^2} \quad \forall p > 0$$

Properties of Laplace

1) Linearity property

$$\mathcal{L}[f(t)] = F(p); \mathcal{L}[g(t)] = G(p)$$

$\forall$ const. a & b ;

$$\mathcal{L}[af(t) + bg(t)] = a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)]$$

eg $\mathcal{L}[\cosh at]$ cos hyperbolic funcⁿ

$$= \mathcal{L}\left[\frac{e^{at} + e^{-at}}{2}\right]$$

$$= \frac{1}{2} \mathcal{L}[e^{at}] + \frac{1}{2} \mathcal{L}[e^{-at}]$$

$$= \frac{1}{2} \frac{1}{p-a} + \frac{1}{2} \frac{1}{p+a}$$

$\forall p > a$

$\forall p > -a$

$$= \frac{1}{2} \frac{2p}{p^2 - a^2} \quad \forall p > |a|$$

$$\boxed{\mathcal{L}[\cosh at] = \frac{p}{p^2 - a^2} \quad \forall p > |a|}$$

$$\mathcal{L}[\sinh at] = \mathcal{L}\left[\frac{e^{at} - e^{-at}}{2}\right]$$

$$= \frac{1}{2} \frac{1}{p-a} - \frac{1}{2} \frac{1}{p+a}$$

$$\boxed{\mathcal{L}[\sinh at] = \frac{a}{p^2 - a^2} \quad \forall p > |a|}$$

II. FIRST SHIFTING TH^M

$$\mathcal{L}[e^{at} f(t)] = F(p-a)$$

Proof:

$$= \int_0^{\infty} e^{at} f(t) e^{-pt} dt$$

$$= \int_0^{\infty} f(t) e^{-(p-a)t} dt$$

p is parameter
& a is const.

$p-a$ is another parameter.

$$= F(x)$$

$$\boxed{L[e^{at} f(t)] = F(p-a)}$$

$$q'' L[e^{4t} \sin t]$$

$$L[\sin t] = \frac{q}{q^2 + p^2} \quad \forall p > 0$$

$$\boxed{L[e^{4t} \sin t] = \frac{q}{(p-4)^2 + 1} \quad \forall p > 0}$$

III CHANGE OF SCALE PROPERTY

$$\text{Then } \boxed{L[f(at)] = \frac{1}{a} F\left(\frac{p}{a}\right)}$$

$$= \int_0^{\infty} f(at) e^{-pt} dt$$

$$x = at$$

$$dx = a dt$$

$$= \left(\frac{1}{a}\right) \int_0^{\infty} f(x) e^{-p\left(\frac{x}{a}\right)} dx$$

$$= \frac{1}{a} \int_0^{\infty} f(x) e^{-\left(\frac{p}{a}\right)x} dx$$

$$F\left(\frac{p}{a}\right) = f \text{ (diff. parameter)}$$

$$= \frac{1}{a} F\left(\frac{p}{a}\right)$$

$$\boxed{L[f(ax)] = \frac{1}{a} F\left(\frac{p}{a}\right)}$$

eg. $\mathcal{L}[f(t)] = \frac{p^2 - p + 1}{p + 1}$

find $\mathcal{L}[f(2t)]$

$$= \left(\frac{1}{2}\right) \left(\frac{p^2 - \frac{p}{2} + 1}{\frac{p}{2} + 1} \right)$$

$$= \frac{1}{4} \left(\frac{p^2 - 2p + 4}{p + 2} \right)$$

IV. MULTIPLICATION PROPERTY

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{dp^n} F(p)$$

V. DIVISION PROPERTY

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_0^\infty F(u) du$$

VI. DIFFⁿ PROPERTY

$$\mathcal{L}[f'(t)] = p \mathcal{L}[f(t)] - f(0)$$

$$\mathcal{L}[f''(t)] = p \mathcal{L}[f'(t)] - f'(0)$$

$$\mathcal{L}[f'''(t)] = p^2 \mathcal{L}[f(t)] - p f(0) - f'(0)$$

$$\mathcal{L}[f^{(n)}(t)] = p^n \mathcal{L}[f(t)] - p^{n-1} f(0) - p^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

VII. INTEGRAL PROPERTY

$$\mathcal{L}\left[\int_0^t f(u) du\right] = \frac{F(p)}{p}$$

Unit Step Funcⁿ (Heaviside fu)

$$u(t-a) = \begin{cases} 1, & t \geq a \\ 0, & \text{otherwise} \end{cases}$$